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MEDICONNECT AI: A FEDERATED LEARNING AND AGENTIC AI FRAMEWORK FOR HEALTHCARE SUPPLY CHAIN MANAGEMENT

A PROJECT REPORT

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ABSTRACT

The lack of coordination and privacy-sensitive data, along with fragmentation, is an inescapable issue in the management of pharmaceutical supply chains of multi-hospital networks because of a lack of efficiency in coordinating activities between stakeholders. Wastage of drugs that were almost out of date, medicine counterfeits, frequent stockouts, and delay in delivery of goods and services are frequent problems that greatly affect patient care and operational performance. This paper introduces MediConnect AI, a federated learning and agentic artificial intelligence system created as a centralized and non-shared-raw-data institutional supply chain management, allowing autonomous, secure, and intelligent supply chain management.The suggested system combines a set of sophisticated AI paradigms, such as Federated Learning (FL), Agentic AI, Generative AI, and Computer Vision, into a single system. The Federated LSTM models are trained in five hospital nodes via the Flower framework with differential privacy on Opacus to guarantee confidentiality of the data. Also, autonomous agents that are driven by large language models can plan activities, including demand prediction, inventory management, automated procurement, anomaly detection, and verification of compliance. The SHAP-based interpretations of model or agent decisions make explainability possible.The experiment on the MIMIC-III and PharmaSim datasets shows that the accuracy of the forecast is 93.6 percent, there is a drop in the number of stockouts by 34.2 percent, wastage by 28.7 percent, and a detection accuracy of 91.3 percent. The system not only guarantees privacy but also enhances supply chain performance to a great extent

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

AI	--	Artificial Intelligence
API	--	Application Programming Interface
CV	--	Computer Vision
DP	--	Differential Privacy
EHR	--	Electronic Health Record
FAISS	--	Facebook AI Similarity Search
FL	--	Federated Learning
GPT	--	Generative Pre-trained Transformer
GPU	--	Graphics Processing Unit
IoT	--	Internet of Things
LIME	--	Local Interpretable Model-Agnostic Explanations
LLM	--	Large Language Model
LSTM	--	Long Short-Term Memory
mAP	--	Mean Average Precision
OCR	--	Optical Character Recognition
PO	--	Purchase Order
RAG	--	Retrieval-Augmented Generation
RNN	--	Recurrent Neural Network
SCM	--	Supply Chain Management
XAI	--	Explainable Artificial Intelligence
YOLO	--	You Only Look Once
σ	--	Noise standard deviation (differential privacy)
W	--	Model weight vector
α,β,γ	--	Weighting factors in agent decision scoring

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

The healthcare supply chain is one of the most complex and critical operational systems in modern medicine. It encompasses the end-to-end flow of pharmaceuticals, medical devices, and consumables from manufacturers and distributors to hospitals, pharmacies, and ultimately patients. Efficient management of this supply chain is essential to ensure the timely availability of medicines and medical supplies, directly impacting patient outcomes and healthcare costs.

In the context of multi-hospital networks, pharmaceutical supply chain management becomes exponentially more challenging. Each hospital operates as a semi-autonomous entity with its own procurement systems, inventory management processes, and supplier relationships. The lack of real-time coordination between these hospitals results in a series of systemic inefficiencies: drug stockouts occur when hospitals underestimate demand; overstocking leads to financial waste and near-expiry drug disposal; counterfeit drug infiltration poses direct risks to patient safety; and procurement delays disrupt the continuity of care.

Traditional approaches to addressing these challenges rely on manual processes, periodic reporting, and siloed information systems that are unable to capture the dynamic nature of pharmaceutical demand. Even where digital systems exist, they typically lack intelligence they record data but do not predict, advise, or act autonomously. The result is an industry that remains heavily dependent on human expertise and reactive decision-making, both of which introduce delays, errors, and inefficiencies.

Artificial Intelligence (AI) has emerged as a transformative technology across industries, and healthcare logistics is no exception. Machine learning models have demonstrated the ability to predict demand with high accuracy by learning from historical consumption patterns, seasonal trends, and contextual factors. However, the deployment of standard AI in multi-hospital pharmaceutical supply chains faces a fundamental contradiction: the data required to train accurate predictive models is distributed across hospitals and is inherently sensitive containing patient treatment records, prescribing patterns, and procurement details making centralized data collection legally and ethically problematic.

This tension between the need for collaborative intelligence and the imperative of data privacy has motivated the development of Federated Learning (FL) a paradigm that enables multiple institutions to jointly train a shared model without ever transmitting raw data. Instead, only model parameters (weights) are exchanged, and differential privacy mechanisms ensure that even these updates cannot be reverse-engineered to reveal sensitive information. Federated Learning thus provides a principled solution to the privacy-performance trade-off in multi-institutional AI systems.

Beyond predictive forecasting, the management of a pharmaceutical supply chain requires continuous monitoring, autonomous decision-making, and real-time responses to supply disruptions. This calls for a paradigm shift from passive prediction to active, autonomous management a capability embodied by Agentic AI systems. Powered by large language models, agentic systems can perceive their environment, reason over complex information, and execute actions across multiple operational domains without continuous human supervision.

The integration of Federated Learning, Agentic AI, Generative AI for intelligent reporting, and Computer Vision for real-time drug verification represents a comprehensive, privacy-preserving, and autonomous solution to the multi-dimensional challenges of pharmaceutical supply chain management.

1.2 Statement of the Problem

Multi-hospital pharmaceutical supply chains face a confluence of structural and operational challenges that cannot be adequately addressed by existing technologies. A systematic analysis reveals four primary problem dimensions:

1.2.1 Data Privacy and Fragmentation

Hospitals in a network generate vast quantities of operationally valuable data — usage histories, stock movements, procurement records, expiry schedules, and patient demand patterns.

However, this data is highly sensitive and subject to strict regulatory frameworks (HIPAA, GDPR, and national healthcare data protection laws).

Centralizing this data for joint analysis is not only legally risky but practically infeasible due to institutional reluctance to share competitive and patient-sensitive information. As a result, each hospital operates in informational isolation, unable to benefit from the collective intelligence of the network.

1.2.2 Reactive and Manual Operations

Current pharmaceutical supply chain management is predominantly reactive. Stock replenishment is triggered by threshold alerts rather than predictive forecasting. Procurement decisions are made manually, introducing delays between demand recognition and order fulfillment. Supplier evaluation is periodic rather than continuous. These manual processes are error-prone, slow, and incapable of adapting to sudden demand spikes such as those caused by disease outbreaks or supply disruptions in real time.

1.2.3 Counterfeit Drug Infiltration

The World Health Organization estimates that up to 10% of medicines in low- and middle-income countries are substandard or counterfeit.

Even in well-regulated environments, counterfeit drugs occasionally infiltrate supply chains through fraudulent suppliers or compromised distribution channels. Current inspection processes rely on manual visual examination, which is slow, inconsistent, and unable to detect sophisticated counterfeits. An automated, AI-powered verification system capable of inspecting drug packages at the point of receipt is critically needed.

1.2.4 Near-Expiry Drug Wastage

Pharmaceutical wastage represents both a financial loss and an ethical concern. Drugs that are procured in excess of demand or not redistributed before their expiry date must be disposed of, resulting in direct financial costs and environmental impacts. In large hospital networks, the scale of this problem is significant: even a 1% reduction in pharmaceutical waste can translate to millions in annual savings. Addressing this requires a proactive, intelligent system that tracks shelf lives in real time and triggers redistribution alerts before drugs reach their expiry threshold.

1.3 Objectives of the Study

In light of the identified problem dimensions, this project defines the following specific objectives:

1. To design and implement a Federated Learning-based demand forecasting system using LSTM networks coordinated via the Flower framework, incorporating Opacus-based differential privacy to ensure data confidentiality across hospital nodes.
2. To develop a multi-agent Agentic AI layer using LangGraph and large language models (GPT-4o, Llama 3 70B) comprising specialized agents for inventory monitoring, shortage prediction, automated procurement, supplier evaluation, expiry tracking, anomaly detection, and compliance checking.

3. To integrate a Generative AI and Retrieval-Augmented Generation (RAG) pipeline using FAISS for intelligent natural language querying, automated supply chain report generation, and a multilingual voice interface using OpenAI Whisper.
4. To implement a real-time drug verification module using YOLOv8 for object detection and PaddleOCR for text extraction, enabling automated authentication of pharmaceutical packages against inventory records.
5. To integrate SHAP-based Explainable AI (XAI) to provide interpretable explanations for model predictions and agent decisions, building trust in the system among healthcare administrators.
6. To evaluate the complete system on the MIMIC-III clinical database and PharmaSim pharmaceutical simulation datasets, demonstrating measurable improvements in forecasting accuracy, stockout reduction, waste minimization, and drug detection performance over baseline approaches.

1.4 Scope and Limitations

The scope of MediConnect AI encompasses the full pharmaceutical supply chain within multi-hospital network environments, from demand forecasting and inventory management through procurement automation, supplier coordination, drug verification, and compliance reporting. The system is designed to be deployable in hospital networks ranging from regional consortia to national healthcare systems.

The current implementation and evaluation are conducted on simulated hospital nodes using the MIMIC-III clinical database and the PharmaSim pharmaceutical simulation dataset. Real-world deployment would require integration with hospital information systems (HIS) and electronic health record (EHR) platforms, regulatory validation under applicable healthcare data protection laws, and clinical validation trials. These activities are identified as future work.

The system assumes reliable network connectivity between hospital nodes and the federated coordination server. Performance under degraded network conditions, which may affect parameter synchronization in resource-constrained environments, has not been evaluated in this study.

1.5 Organization of the Report

The remainder of this report is organized as follows. Chapter 2 provides a comprehensive review of relevant literature across federated learning, AI in healthcare supply chains, agentic AI systems, computer vision, and explainable AI, identifying key contributions and research gaps. Chapter 3 describes the proposed MediConnect AI system architecture, presenting each of its five major modules in detail. Chapter 4 details the methodology employed, including data handling, model training formulations, agent workflow design, and drug verification pipeline. Chapter 5 presents experimental results with detailed analysis and discussion. Chapter 6 concludes the report with a summary of contributions and directions for future work. References and appendices follow.

CHAPTER 2

LITERATURE REVIEW

2.1 Federated Learning in Healthcare

The concept of Federated Learning (FL) was formalized by McMahan et al. [4] with the introduction of the Federated Averaging (FedAvg) algorithm. FedAvg enables distributed clients to collaboratively train a shared global model by exchanging only model parameters rather than raw data. Each client trains a local model on its own dataset and transmits the resulting model weights to a central aggregation server, which computes a weighted average to produce the global model. This process iterates over multiple communication rounds until convergence. The mathematical foundation of FedAvg specifically the weighted aggregation formula ,

$$\mathbf{W_global = \Sigma(n_i/n_total)\cdot W_i}$$

has become the standard basis for federated model training. Rieke et al. [1] conducted a seminal study on the future of digital health with federated learning, demonstrating that FL can achieve model performance comparable to centralized training while fully preserving data privacy. Their work established theoretical bounds on privacy guarantees and convergence rates in heterogeneous healthcare data settings. Sheller et al. [2] applied FL specifically to medical imaging analysis, conducting a feasibility study on brain tumor segmentation across multiple institutions. Their results showed that federated models trained without sharing patient data could match the performance of models trained on pooled datasets, validating FL as a practical approach for sensitive healthcare applications.

Li et al. [3] addressed a key limitation of FedAvg — its sensitivity to statistical heterogeneity across clients — by proposing FedProx, a federated optimization algorithm that adds a proximal term to the local optimization objective.

FedProx demonstrated improved convergence stability and model accuracy in scenarios where client data distributions differ significantly, a condition that is pervasive in multi-hospital pharmaceutical datasets where patient demographics, disease profiles, and prescribing practices vary substantially between institutions.

From a privacy standpoint, the application of differential privacy (DP) to federated learning has been extensively studied. Differential privacy, formalized by Dwork et al., provides a mathematical guarantee that the inclusion or exclusion of any individual data point does not significantly affect the output of an algorithm. In the context of FL, DP is implemented by adding calibrated Gaussian noise to model gradients before transmission:

$$\tilde{\mathbf{g}} = \mathbf{g} + \mathbf{N}(\mathbf{0}, \boldsymbol{\sigma}^2),$$

where σ is chosen to satisfy a specified privacy budget (ε, δ). The Opacus library from Meta provides a practical PyTorch implementation of DP-SGD that has been widely adopted in privacy-preserving ML applications.

Despite the substantial body of work on FL in healthcare, existing applications have focused primarily on clinical prediction tasks (disease diagnosis, mortality prediction, imaging analysis) rather than operational supply chain management. The application of FL to pharmaceutical demand forecasting in multi-hospital networks represents a significant and underexplored contribution of this project.

Table 2.1: Summary of federated learning approaches in healthcare

Authors	Method	Application Domain	Key Finding
McMahan et al. [4] (2017)	FedAvg	Distributed ML	Foundation FL algorithm
Rieke et al. [1] (2020)	FL Survey	Digital Health	FL viable for sensitive clinical data
Sheller et al. [2] (2019)	Multi-Inst. FL	Brain Tumor Seg.	Matches centralized performance
Li et al. [3] (2020)	FedProx	Heterogeneous FL	Improved convergence in non-IID data

2.2 Deep Learning for Healthcare Analytics

Deep learning has achieved remarkable results in healthcare analytics, particularly for tasks involving sequential data and complex pattern recognition. Rajkomar et al. [5] demonstrated that deep neural networks trained on Electronic Health Record (EHR) data can predict patient outcomes including in-hospital mortality, 30-day readmission, and prolonged length of stay — with performance exceeding that of traditional clinical scoring systems. Their work established the feasibility of deploying large-scale deep learning in healthcare settings and highlighted the importance of EHR-derived features, many of which directly inform pharmaceutical demand patterns.

Esteva et al. [6] achieved dermatologist-level classification of skin cancer using convolutional neural networks (CNNs), demonstrating that deep learning can match or exceed human expert performance in medical image analysis.

While this work focused on dermatology, the underlying finding that deep neural networks can extract clinically meaningful features from high-dimensional data is broadly applicable to pharmaceutical image recognition tasks such as drug package verification.Johnson et al. [7] introduced the MIMIC-III (Medical Information Mart for Intensive Care III) clinical database, a freely accessible repository of electronic health records for over 40,000 critical care patients. MIMIC-III has become the de facto benchmark dataset for healthcare analytics research, providing a rich source of time-series clinical data that can be used to simulate realistic pharmaceutical demand patterns across hospital networks. This project uses MIMIC-III as one of its primary evaluation datasets.

For time-series forecasting specifically, Long Short-Term Memory (LSTM) networks have demonstrated superior performance over traditional statistical models such as ARIMA and exponential smoothing. LSTMs address the vanishing gradient problem inherent in standard recurrent neural networks (RNNs) through a gated architecture that selectively retains or discards information over long time horizons. This capability is particularly valuable for pharmaceutical demand forecasting, where demand patterns exhibit complex temporal dependencies including weekly seasonality, monthly procurement cycles, and event-driven spikes.

2.3 AI in Supply Chain Management

The application of artificial intelligence to supply chain management has accelerated significantly over the past decade, driven by the availability of large-scale operational data and advances in machine learning. Ivanov et al. [8] provided a comprehensive analysis of how digital technologies, including AI, IoT, and blockchain, are transforming supply chain resilience and risk management. Their analysis of Industry 4.0 impacts on supply chain disruption highlighted the critical role of predictive analytics and autonomous decision-making in managing ripple effects the cascading failures that propagate through interconnected supply chains following an initial disruption.

In the pharmaceutical domain specifically, Kwon et al. [9] developed predictive analytics models for drug demand forecasting. Their work demonstrated that machine learning models, particularly those incorporating contextual features such as disease prevalence data and demographic trends, substantially outperform naive demand extrapolation approaches in predicting drug consumption. The reduction in both stockouts and overstocking achieved through ML-based forecasting translates directly to improved patient care outcomes and reduced operational costs.

Dolgui et al. [10] introduced the concept of digital supply chain twins virtual replicas of physical supply chain systems that enable real-time simulation, monitoring, and optimization. Digital twins provide a unified operational view across supply chain nodes, facilitating faster identification of disruptions and enabling what-if analysis for procurement decisions.MediConnect AI incorporates elements of this concept through its federated monitoring architecture, which provides a synchronized view of inventory states across all participating hospital nodes without centralizing sensitive data.

The limitations of existing supply chain AI systems are primarily two-fold. First, they operate on centralized data, making them unsuitable for privacy-sensitive multi-hospital environments. Second, they function as decision support tools rather than autonomous agents — they provide recommendations but require human intervention to execute decisions. MediConnect AI addresses both limitations through its federated architecture and agentic AI layer.

2.4 Agentic AI and Large Language Models

The concept of agentic AI — systems that autonomously perceive their environment, reason over information, and execute actions to achieve specified goals — has gained substantial momentum following the rapid advancement of large language models. Park et al. [11] introduced Generative Agents, a framework in which LLM-powered agents simulate complex human-like behaviors in open-ended environments.

Their seminal demonstration showed that agents equipped with memory, reflection, and planning capabilities can coordinate, adapt, and exhibit emergent social behaviors, providing a proof of concept for LLM-based autonomous systems in complex operational environments.

Yao et al. [12] proposed ReAct (Reasoning + Acting), a paradigm that synergizes reasoning traces and action execution in language models. ReAct agents interleave verbal reasoning steps with calls to external tools (databases, APIs, calculators), enabling them to solve complex multi-step tasks that require both knowledge retrieval and decision execution. The ReAct framework directly informs the design of MediConnect AI's autonomous agents, which must reason over inventory data, demand forecasts, supplier records, and compliance policies to generate procurement decisions.

The release of GPT-4 [13] by OpenAI marked a qualitative advance in LLM capabilities, demonstrating strong performance on complex reasoning, code generation, and structured planning tasks. The subsequent development of the LLaMA family of open-source models by Touvron et al. [14] provided a viable on-premise alternative to cloud-based LLMs, enabling deployment of powerful language models in resource-constrained or privacy-sensitive environments. MediConnect AI uses GPT-4o as the primary reasoning engine for cloud-connected deployments and Llama 3 70B for on-premise configurations, providing flexibility across different institutional infrastructure models.

LangGraph's graph-based approach provides the structured coordination mechanism required for hierarchical agent architecture, ensuring that decisions made by lower-level monitoring agents are correctly escalated to and acted upon by higher-level procurement and compliance agents.

2.5 Computer Vision in Pharmaceutical Applications

Computer vision has been applied to pharmaceutical quality control and drug verification across a range of contexts. The You Only Look Once (YOLO) family of real-time object detection models, introduced by Redmon et al. [17], revolutionized the field by achieving single-pass detection of multiple objects with performance sufficient for real-time applications. The original YOLO architecture was subsequently refined through multiple iterations, culminating in YOLOv8 [15], which provides state-of-the-art object detection performance with improved efficiency for deployment on edge devices.

YOLOv8 achieves this through an anchor-free detection head, a C2f (Cross Stage Partial) backbone for improved feature extraction, and a decoupled head design that separates classification and localization tasks. For pharmaceutical package detection, YOLOv8 can identify medicine boxes, labels, barcodes, and serial number regions from camera images with high speed and accuracy, providing the foundational detection capability for MediConnect AI's drug verification module.

Optical Character Recognition (OCR) technology enables the extraction of structured text from images, a critical capability for reading drug names, batch numbers, manufacturing and expiry dates from pharmaceutical labels. PaddleOCR [16], developed by Baidu, is a lightweight and practical OCR system that supports multilingual text recognition . PaddleOCR's architecture combines a detection model (based on DBNet) with a recognition model (based on SVTR) to provide accurate end-to-end text extraction from complex label images. The combination of YOLOv8 for region detection and PaddleOCR for text recognition provides a complete pipeline for pharmaceutical package verification: detecting the label region, extracting all printed text fields, and cross-referencing the extracted data against expected inventory records. This pipeline enables automated detection of counterfeit drugs, mislabeled products, and near-expiry items at the point of receipt, significantly reducing the risk of compromised drugs entering clinical use.

2.6 Explainable AI and Retrieval-Augmented Generation

The adoption of AI in high-stakes healthcare applications is contingent on the ability to explain model predictions to domain experts and regulators. Explainable AI (XAI) provides methods for interpreting the decisions of complex machine learning models in terms that are meaningful to human users. Lundberg and Lee [18] introduced SHAP (SHapley Additive exPlanations), a unified framework for model interpretability based on Shapley values from cooperative game theory. SHAP provides consistent, locally accurate feature attribution explanations for any machine learning model, making it widely applicable across the diverse model types used in MediConnect AI — from LSTM forecasting models to LLM-based agent decisions.Ribeiro et al. [19] proposed LIME (Local Interpretable Model-Agnostic Explanations), which approximates complex models locally with interpretable surrogate models. While LIME provides useful local explanations, SHAP is preferred in MediConnect AI due to its superior theoretical consistency and its ability to capture both local and global feature importance.

Together, SHAP and LIME represent the state of the art in model-agnostic XAI, enabling MediConnect AI to provide transparent explanations for every demand forecast and procurement recommendation it generates.Lewis et al. [20] introduced Retrieval-Augmented Generation (RAG), a framework that enhances generative language models with access to external knowledge repositories.

2.7 Research Gaps and Motivation

The review of existing literature reveals a clear and significant gap: while federated learning, agentic AI, computer vision, and explainable AI have each been extensively studied in isolation, no existing system integrates all of these technologies into a unified, end-to-end pharmaceutical supply chain management framework. Existing federated learning systems in healthcare focus on clinical prediction rather than operational logistics.

Existing supply chain AI systems rely on centralized data and lack autonomous decision-making. Existing drug verification systems are either manual or employ simple threshold-based image processing.

MediConnect AI addresses this gap by providing the first comprehensive, privacy-preserving, autonomous, and explainable AI framework for multi-hospital pharmaceutical supply chain management. The integration of federated demand forecasting, multi-agent autonomous operations, generative AI knowledge assistance, and real-time computer vision drug verification within a single coherent architecture represents a genuinely novel contribution to both the AI and healthcare informatics communities.

CHAPTER 3

PROPOSED SYSTEM

3.1 System Overview

MediConnect AI is a comprehensive, privacy-preserving, and autonomous healthcare pharmaceutical supply chain management system. It is designed to address the four primary challenge dimensions identified in Chapter 1: data privacy and fragmentation, reactive and manual operations, counterfeit drug infiltration, and near-expiry drug wastage. The system achieves this through the integration of five major technical modules: (1) Federated Demand Forecasting, (2) Agentic Autonomous AI Layer, (3) Generative AI and Knowledge Assistance, (4) Computer Vision Drug Verification, and (5) a unified System Architecture that integrates all modules into a coherent operational platform.

The foundational design principle of MediConnect AI is that collective intelligence can be achieved without compromising individual privacy. By keeping raw data at its point of origin and sharing only encrypted, differentially private model parameters, the system enables the five participating hospital nodes to benefit from the combined predictive power of the entire network while each hospital retains full control over its sensitive operational data. This privacy-by-design approach is not merely a technical feature but a fundamental architectural commitment that makes MediConnect AI deployable in real-world healthcare environments. Autonomy is the second foundational principle. The system is designed to minimize the requirement for human intervention in routine supply chain decisions. By deploying a hierarchy of specialized AI agents that continuously monitor inventory states, generate demand forecasts, identify shortage risks, create procurement orders, evaluate suppliers, track expiry dates, detect anomalies, and verify compliance to all without manual initiation .

MediConnect AI enables supply chain managers to shift their focus from routine operational tasks to strategic decision-making and exception handling.

3.2 Federated Demand Forecasting Module

The Federated Demand Forecasting module is the predictive core of MediConnect AI. Its purpose is to forecast future pharmaceutical demand across all hospital nodes with high accuracy, while ensuring that the sensitive consumption and operational data that underpins these forecasts never leaves its institutional origin.

3.2.1 Data Localization Architecture

Each hospital node maintains a local data repository containing: (a) daily medicine consumption records, including quantities dispensed per drug per ward; (b) real-time inventory levels for all stocked pharmaceuticals; (c) historical procurement records, including order quantities, supplier identities, delivery lead times, and unit prices; (d) drug expiry schedules, tracking the manufacturing date, expiry date, and remaining shelf life of every batch in stock; and (e) relevant contextual data including hospital occupancy rates and seasonal disease incidence patterns. This data is organized into time-series sequences suitable for LSTM model training. For each pharmaceutical, a sliding window of the past T timesteps (typically T = 30 days) is used as the input feature vector, with the consumption on day T+1 as the prediction target. The data remains entirely within the hospital node and is never transmitted to any external system. Only the trained model parameters specifically the LSTM weight matrices are extracted and transmitted, under encryption, to the federated coordination server.

3.2.2 LSTM Architecture

The demand forecasting model at each hospital node is an LSTM network with the following architecture: an input layer accepting the time-series feature vector, two LSTM layers with 128 hidden units each and dropout regularization ($p = 0.2$) to prevent overfitting, a fully connected (dense) output layer producing a scalar demand forecast, and a final ReLU activation to ensure non-negative demand predictions.

The model is trained using the Adam optimizer with an initial learning rate of 0.001, mean squared error (MSE) loss function, and early stopping based on validation loss with a patience of 10 epochs.

3.2.3 Federated Coordination via Flower

The Flower (flwr) federated learning framework coordinates the distributed training process across all hospital nodes. Flower provides a client-server architecture in which the federated server orchestrates training rounds and each hospital node operates as a Flower client.

In each training round: (1) the server distributes the current global model to all clients; (2) each client trains the model on its local data for a specified number of local epochs; (3) each client serializes and uploads its updated model parameters; (4) the server aggregates the uploaded parameters using the FedAvg algorithm; and (5) the server distributes the updated global model to all clients for the next round. This process continues for a specified number of global rounds until the global model converges.

3.3 Agentic Autonomous AI Layer

The Agentic AI Layer constitutes the operational intelligence of MediConnect AI. It comprises a hierarchy of specialized autonomous software agents, each responsible for a specific supply chain management function. These agents are implemented using the LangGraph framework and powered by either GPT-4o (for cloud-connected deployments) or Llama 3 70B (for on-premise deployments).

The agents operate continuously, processing inputs from the forecasting module, the inventory database, supplier records, and regulatory compliance rules to generate and execute supply chain decisions.

Table 3.1: Agent roles and responsibilities in the agentic AI layer

Agent Name	Primary Responsibility	Key Inputs / Outputs
Inventory Surveillance	Monitor stock levels in real time	Inventory DB → alerts, dashboards
Shortage Prediction	Compute time-to-stockout per drug	LSTM forecast + inventory → shortage alerts
Purchase Order Agent	Auto-generate POs on threshold breach	Shortage alert → structured PO
Supplier Negotiation	Evaluate and rank suppliers	PO + supplier DB → optimal supplier selection
Expiry Tracking	Flag near-expiry batches	Inventory + expiry dates → redistribution alerts
Anomaly Detection	Detect unusual demand or supply events	Forecast + actual usage → anomaly flags
Compliance Checking	Verify regulatory conformance	PO + regulatory rules → compliance status
Risk Scoring	Score supply chain risks dynamically	All agent outputs → composite risk score

3.3.1 Decision Scoring Mechanism

To prioritize competing actions across multiple agents, MediConnect AI employs a quantitative decision scoring mechanism. Each potential supply chain action is assigned a composite score based on three weighted factors:

$$\mathbf{Score = \alpha \times Demand + \beta \times Risk + \gamma \times SupplierReliability}$$

Where α, β, and γ are configurable weighting factors that allow system administrators to tune the prioritization logic according to institutional preferences.

The Demand factor reflects the urgency of the supply requirement (derived from the LSTM forecast and current inventory level). The Risk factor captures the potential impact of a supply failure for the specific drug category. The SupplierReliability factor reflects the historical performance of the selected supplier in terms of on-time delivery and product quality. Actions with the highest composite scores are executed first, ensuring that critical drug procurement needs are addressed before lower-priority replenishment tasks.

3.4 Generative AI and Knowledge Assistance

The Generative AI module extends MediConnect AI beyond autonomous operations to provide intelligent knowledge assistance to human users — administrators, pharmacists, procurement managers, and supply chain analysts — who require access to supply chain information and decision support.

3.4.1 FAISS-based RAG Pipeline

A comprehensive knowledge base is constructed by indexing hospital supply chain documents policies, supplier contracts, inventory logs, procurement records, compliance guidelines, and historical queries — into a FAISS vector store using dense text embeddings. When a user submits a natural language query, the system performs a semantic similarity search over the indexed documents to retrieve the most relevant context passages. These passages are then provided as context to GPT-4o, which generates a coherent, contextually grounded response.

3.4.2 Automated Report Generation

The Generative AI module automatically produces structured supply chain reports on a scheduled basis: weekly supply chain performance summaries, shortage risk reports, compliance audit documentation, and supplier performance evaluations.

These reports are generated by prompting GPT-4o with structured data outputs from the agent layer and the forecasting module, combined with institution-specific templates retrieved from the knowledge base via RAG. The result is professional, readable documentation that requires no manual preparation and is available to stakeholders immediately upon completion of each operational cycle.

3.4.3 Multilingual Voice Interface

To maximize accessibility in diverse healthcare environments particularly in India where hospital staff may be most comfortable communicating in regional languages .MediConnect AI incorporates a multilingual voice interface based on OpenAI Whisper (large-v3). Users can submit voice queries in Tamil, Hindi, or English, which Whisper transcribes to text before passing to the RAG pipeline. The system's text responses are optionally converted back to speech using a text-to-speech synthesizer, enabling fully voice-driven interaction with the supply chain management system.

3.5 Drug Verification through Computer Vision

The Computer Vision Drug Verification module provides automated, real-time authentication of pharmaceutical packages at the point of receipt. This module directly addresses the challenge of counterfeit drug infiltration and provides a scalable alternative to manual visual inspection.

3.5.1 YOLOv8 Object Detection

When a drug shipment is received, the hospital staff scans packages using either a fixed camera station or a mobile device running the MediConnect AI application. The camera feed is processed in real time by YOLOv8, which detects and localizes all relevant regions of the drug package: the primary label (containing drug name, batch number, dates, and dosage information), the barcode or QR code, and any secondary labels (storage instructions, warnings, regulatory markings). YOLOv8 returns bounding box coordinates and confidence scores for each detected region.

3.5.2 PaddleOCR Text Extraction

Each detected label region is cropped from the original image and passed to PaddleOCR for text extraction. PaddleOCR's detection model identifies text line regions within the cropped image, and the recognition model transcribes each line to digital text. The extracted text fields — drug name, batch number, manufacturing date, expiry date, dosage strength, and barcode value — are structured into a machine-readable record for comparison against the expected inventory.

3.5.3 Verification and Alerting

The extracted drug record is compared against the purchase order and expected inventory record in the MediConnect AI database.

Discrepancies in drug name, batch number, expiry date, or manufacturer identity are flagged as potential counterfeits or labeling errors. Items with expiry dates within a configurable threshold (e.g., 90 days) are flagged for priority usage or return to supplier. Items whose batch numbers do not match any expected delivery are escalated to the compliance checking agent for further investigation. All verification outcomes are logged for audit and compliance reporting purposes.

3.6 System Architecture and Workflow

The complete MediConnect AI system is organized as a multi-layered client-server architecture comprising four tiers: Hospital Nodes, Federated Coordination Server, AI Coordination Layer, and Frontend User Interface. Figure 3.1 provides a visual overview of the system architecture.

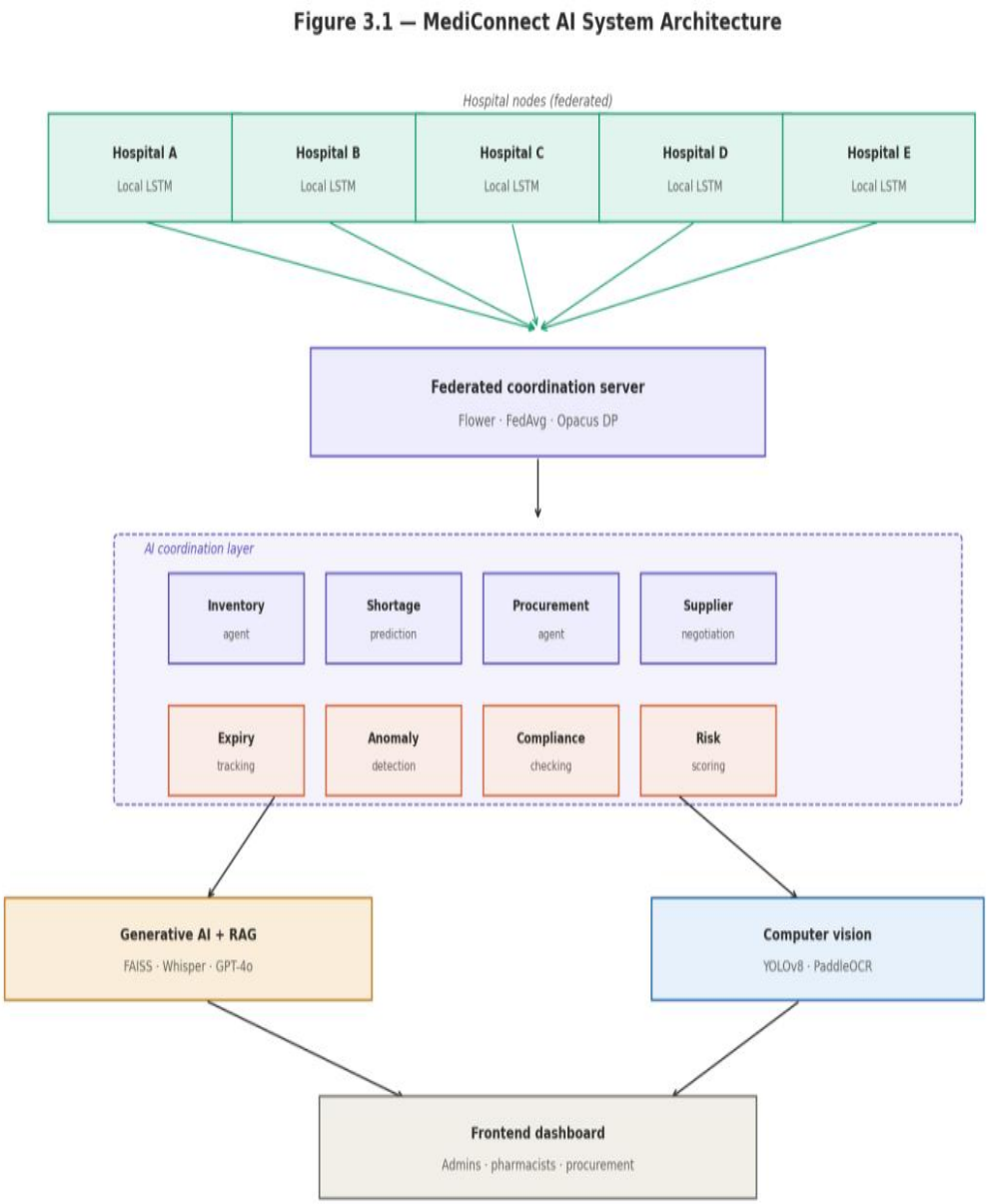


Figure 3.1: System architecture of MediConnect AI

At the bottom tier, five hospital nodes each operate their own local LSTM model on their own private data. The federated coordination server are the second tier orchestrates the training process using Flower, collecting encrypted model parameter updates from each node, computing the federated average, and redistributing the updated global model.

The AI Coordination Layer is the third tier that houses all eight autonomous agents and the Generative AI pipeline, operating continuously on the aggregated intelligence provided by the global forecasting model. The Frontend is the fourth tier that provides role-specific dashboards for hospital administrators, pharmacists, and procurement teams, as well as the voice interface and computer vision scanning capability.

The workflow of a typical operational cycle proceeds as follows: (1) Each hospital node trains its local LSTM on the most recent local data; (2) Updated parameters are transmitted to the federated server and aggregated; (3) The global model is distributed back to all nodes; (4) The inventory surveillance agent reads current stock levels and compares them against the updated demand forecast; (5) Where deficits are projected, the shortage prediction agent calculates time-to-stockout; (6) The purchase order agent creates procurement requests for critical items; (7) The supplier negotiation agent evaluates and selects optimal suppliers; (8) Approved orders are dispatched electronically to suppliers; (9) Upon receipt, the computer vision module verifies all incoming drug packages; (10) The generative AI module compiles and distributes the weekly supply chain performance report.

CHAPTER 4

METHODOLOGY

4.1 Federated Data Handling and Data Acquisition

The methodology begins with a structured data acquisition process at each of the five simulated hospital nodes. In the real-world deployment scenario, this data exists naturally within hospital information systems and pharmacy management systems. For the purposes of this evaluation, pharmaceutical consumption data is derived from the MIMIC-III clinical database, which provides time-series records of medication administration for over 40,000 ICU patients across two hospitals. Drug utilization patterns, dosage information, and administration frequencies are extracted from the MIMIC-III PRESCRIPTIONS and INPUTEVENTS tables and aggregated to daily consumption totals per drug category.

Inventory level data is simulated using the PharmaSim dataset, a purpose-built pharmaceutical supply chain simulation that models stock movements, procurement events, supplier lead times, and expiry schedules across multiple hospital nodes with heterogeneous demand profiles. The combination of MIMIC-III for realistic consumption patterns and PharmaSim for supply chain operational context provides a comprehensive and realistic evaluation environment.

At each hospital node, the local dataset is stored in a structured format with the following fields per record: (i) date, (ii) drug identifier (ATC code and brand name), (iii) daily consumption quantity, (iv) current inventory level, (v) days of supply remaining (inventory / average daily consumption), (vi) next scheduled procurement date, (vii) nearest expiry date, (viii) number of batches with expiry within 90 days, and (ix) a binary stockout indicator (0 = sufficient stock, 1 = stockout occurred). This structured schema facilitates both LSTM time-series model training and agent-based inventory monitoring.

4.2 Data Preprocessing and Feature Engineering

Raw pharmaceutical data is subject to a range of quality issues — missing records due to system outages, reporting delays, and transcription errors — that must be addressed before model training. The preprocessing pipeline applies the following steps in sequence:

7. Missing value imputation: Missing daily consumption values (representing less than 2% of records across all nodes) are imputed using forward-fill (carry-forward of the most recent valid observation) for consecutive gaps of up to 3 days, and linear interpolation for longer gaps.
8. Outlier detection and treatment: Demand values more than three standard deviations above the 90-day rolling mean are identified as potential data entry errors or exceptional demand events. These values are flagged and capped at the 99th percentile to prevent them from dominating the loss function during model training.
9. Normalization: All numerical features are scaled to the range [0, 1] using Min-Max normalization: $x' = (x - x_{min}) / (x_{max} - x_{min})$. Normalization ensures that features with large absolute values (e.g., high-volume drug quantities) do not dominate features with small absolute values (e.g., days of supply) during LSTM gradient updates.
10. Sequence construction: The normalized time series are segmented into sliding windows of length $T = 30$ days with a single-step forecast horizon. Each window consists of 30 consecutive daily records as input features and the consumption value on day 31 as the prediction target. Overlapping windows are generated with a stride of 1 day, maximizing the training data volume.

Table 4.1: Key features extracted during preprocessing

Feature	Type	Rationale
Daily consumption (normalized)	Continuous	Primary target and key input
7-day moving average	Derived	Captures short-term trend
30-day moving average	Derived	Captures medium-term seasonality
Day of week (one-hot)	Categorical	Weekly demand cycles
Month of year (one-hot)	Categorical	Seasonal demand patterns
Days of supply remaining	Continuous	Direct inventory risk indicator
Expiry risk score	Derived	Near-expiry fraction of stock
Lead time (last procurement)	Continuous	Procurement timing feature

4.3 Federated LSTM-based Demand Forecasting

At each hospital node, an LSTM model is trained on the local preprocessed dataset. The LSTM architecture consists of two stacked LSTM layers, each with 128 hidden units, followed by a dropout layer (dropout rate 0.2) and a fully connected output layer.

The two-layer architecture enables the model to capture both short-term (day-to-day fluctuation) and longer-term (weekly, seasonal) patterns in pharmaceutical demand.Local training at each node proceeds for 20 local epochs per federated round, using mini-batches of 32 sequences. The Adam optimizer with a learning rate of 0.001 is used, with ReduceLROnPlateau scheduling that halves the learning rate when validation loss fails to improve for 5 consecutive epochs. A 80/10/10 train/validation/test split is applied within each node's local dataset.

After completing local training, each node computes the model weight update as the difference between the post-training and pre-training parameter vectors: $\Delta W_i = W_i_post - W_i_pre$. This update vector is compressed using sparsification (zeroing out updates below a magnitude threshold of 0.001) to reduce communication overhead, then encrypted using standard TLS before transmission to the federated coordination server.The federated coordination server runs the FedAvg aggregation algorithm. Given weight updates from N participating hospital nodes, the global model update is computed as:

$$\mathbf{W_global = \Sigma_{i=1}^{N} (n_i / n_total) \times W_i}$$

Where W_i represents the post-training model weights at hospital node i, n_i is the number of training samples at node i, and $n_total = \Sigma n_i$ is the total number of training samples across all nodes. This weighted average ensures that hospitals with larger training datasets exert proportionally greater influence on the global model, reflecting the relative richness of their data contribution to the federation.The complete federated learning workflow is depicted in Figure 4.1.

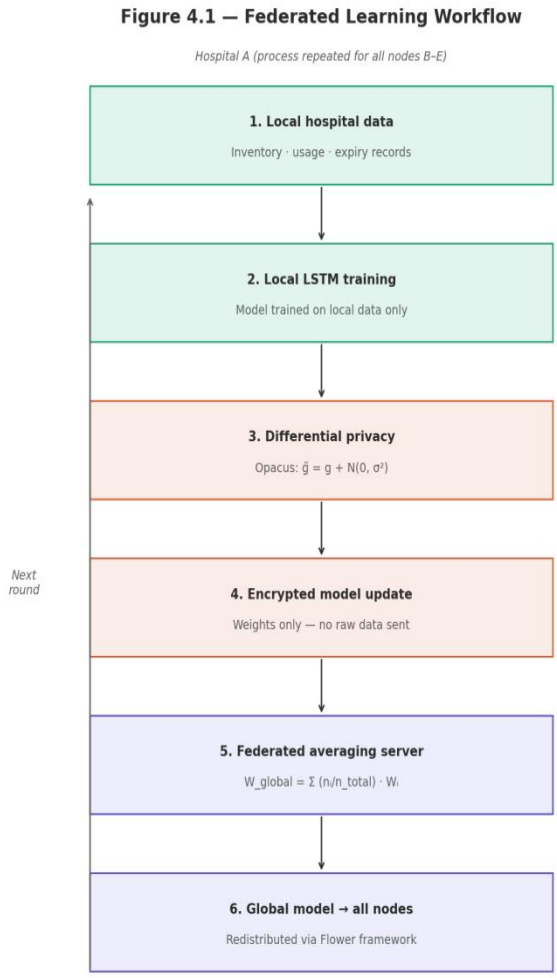


Figure 4.1: Federated learning workflow — from local training to global model aggregation

4.4 Differential Privacy Mechanism

Differential Privacy (DP) provides a formal mathematical guarantee that the trained model reveals minimal information about any individual data point in the training set . In MediConnect AI, DP is applied at the gradient level during local LSTM training using the Opacus library.The DP-SGD (Differentially Private Stochastic Gradient Descent) algorithm operates as follows. For each mini-batch, individual per-sample gradients are computed (rather than the aggregated batch gradient).

Each per-sample gradient is clipped to a maximum L2 norm C: $g_clipped = g \times \min(1, C / \|g\|_2)$. Gaussian noise calibrated to the clipping bound is added to the aggregated gradient: $\tilde{g} = (1/B) \times \Sigma g_clipped + N(0, \sigma^2 C^2 I)$, where B is the batch size and σ is the noise multiplier. The noisy gradient $\tilde{g}$ is then used to update the model parameters via the Adam update rule.The privacy budget is tracked throughout training using the Rényi Differential Privacy (RDP) accountant provided by Opacus. For the experiments reported in this study, DP parameters are set to $\varepsilon = 3.0$, $\delta = 10^{-5}$, $C = 1.0$, and $\sigma = 1.1$. These values provide meaningful privacy protection — an attacker with full access to the model would have at most a probability ratio of $e^3 \approx 20.1$ between any two adjacent datasets in distinguishing their membership — while incurring a forecasting accuracy penalty of less than 0.8% relative to the non-private baseline, as demonstrated in the experimental results.

4.5 Agentic AI Decision-Making Workflow

The agentic AI decision-making workflow coordinates the actions of the eight specialized agents in response to inputs from the forecasting module and the inventory database. The workflow is implemented as a directed graph in LangGraph, with each agent represented as a node and conditional edges representing the routing logic between agents.The workflow initiates at the beginning of each operational cycle (configurable: hourly, daily, or at user request) when the Inventory Surveillance Agent retrieves current stock levels from the inventory database and compares them against the demand forecast for the next 30 days. For each drug where the projected demand exceeds the available supply within a configurable risk horizon (default: 14 days),. Shortage Prediction Agent is invoked to compute the specific time-to-stockout and classify the urgency level (Critical: < 3 days; High: 3–7 days; Medium: 7–14 days).

For Critical and High urgency shortages, the Purchase Order Agent is activated to generate a structured procurement request.

The agent uses an LLM (GPT-4o or Llama 3 70B) to generate the PO content in a structured format, incorporating the required quantity, the preferred delivery window, and any special handling requirements for controlled or temperature-sensitive medications.The generated PO is passed to the Supplier Negotiation Agent, which retrieves the supplier database and computes a composite score for each approved supplier using the decision scoring formula: Score = α × Demand + β × Risk + γ × SupplierReliability. Default weighting values are α = 0.4, β = 0.35, γ = 0.25, placing the highest weight on demand urgency, with significant weight on supply risk and supplier reliability. The supplier with the highest composite score is selected for the order.The complete agentic AI decision-making flow is illustrated in Figure 4.2.

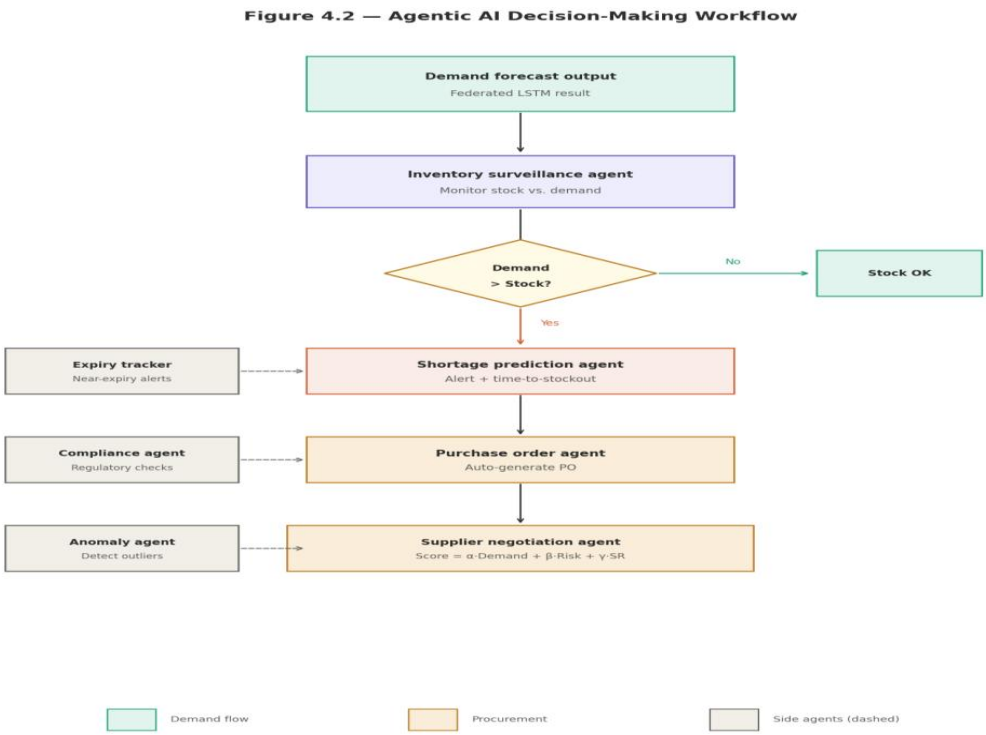


Figure 4.2: Agentic AI decision-making workflow

Parallel to the procurement workflow, the Expiry Tracking Agent continuously monitors the expiry schedules of all stocked pharmaceuticals. Items approaching expiry within the configured threshold are flagged for priority usage these items are assigned priority dispensing instructions so that they are issued before newer stock.

Items within 30 days of expiry that cannot be consumed before their expiry date based on current consumption rate projections are escalated to the Compliance Agent, which initiates a return-to-supplier or inter-hospital redistribution process.The Anomaly Detection Agent operates in parallel with the other agents, monitoring for unusual patterns in demand or supply that may indicate data quality issues, fraudulent activity, or external disruptions. The agent uses a statistical process control (SPC) approach — flagging data points that fall outside three standard deviations of the rolling 90-day distribution — combined with LLM-based contextual reasoning to distinguish genuine anomalies from legitimate demand spikes (e.g., due to disease outbreaks or clinical trial enrollments).

4.6 Automated Drug Verification by Computer Vision

The Computer Vision Drug Verification methodology is designed to provide a fast, reliable, and non-invasive method for authenticating incoming pharmaceutical shipments. The pipeline operates in four stages: image acquisition, object detection, text extraction, and verification.

Image Acquisition: Drug packages are placed on a scanning station equipped with a standardized camera setup — a 12-megapixel camera at a fixed distance of 30 cm from the package surface, with uniform LED lighting to minimize shadows and reflections. Alternatively, the mobile MediConnect AI application supports handheld scanning using smartphone cameras, with image quality assessment to flag and re-request blurred or overexposed captures.

Object Detection via YOLOv8: The acquired image is passed to a YOLOv8 model fine-tuned on a pharmaceutical packaging dataset containing 5,000 annotated images of medicine boxes from 200 manufacturers across India, the UK, and the US. The model detects and localizes four object classes: primary label, barcode/QR code, secondary label, and tamper-evident seal.

For each detected region, the model outputs a bounding box (x, y, w, h), a class label, and a confidence score. Detections with confidence below 0.7 are discarded.

Text Extraction via PaddleOCR: Each detected label region (bounding box) is cropped from the original image and processed by PaddleOCR's two-stage pipeline. The DBNet detection model identifies individual text line regions within the cropped label image. The SVTR recognition model transcribes each text line, handling multilingual labels containing English, Hindi, or Tamil text. The extracted text is post-processed using regular expression patterns to extract structured fields: drug name, batch number, manufacturing date (format: DD/MM/YYYY), expiry date (format: DD/MM/YYYY), dosage strength (format: Xmg or Xmg/mL), and manufacturer name.

Verification: The extracted structured record is compared against the corresponding purchase order and expected delivery record in the MediConnect AI database. The verification logic checks: (a) drug name match (exact string match with fuzzy tolerance ≥ 0.85 using Levenshtein distance to handle OCR errors); (b) batch number match (exact match required); (c) expiry date validity (must be at least 90 days in the future); (d) manufacturer identity (must match approved supplier list). Any discrepancy triggers an alert classified as either REJECT (batch number mismatch or unrecognized manufacturer) or FLAG (expiry within threshold or minor label discrepancy requiring pharmacist review).

4.7 Explainable AI Integration

Explainability is a first-class concern in MediConnect AI, given the high-stakes nature of pharmaceutical supply chain decisions. The system integrates SHAP (SHapley Additive exPlanations) to provide transparent, feature-level explanations for both the LSTM demand forecasting model and the agent decision scoring mechanism. For the LSTM forecasting model, SHAP DeepExplainer is used to compute Shapley values for each input feature in the 30-day sliding window.

The Shapley value for feature j in prediction i quantifies the average marginal contribution of feature j to the prediction, computed across all possible coalitions of features. A positive Shapley value indicates that feature j increased the predicted demand above the baseline (average prediction), while a negative value indicates that it decreased demand below the baseline.These explanations are included in the weekly supply chain reports and are available on-demand through the RAG interface.

Table 4.2: Technology stack and tools used in MediConnect AI

Component	**Technology / Tool**	**Version**	**Purpose**
Federated Learning	Flower (flwr)	1.4.0	FL coordination framework
Privacy Engine	Opacus	1.4.0	Differential privacy for PyTorch
Forecasting Model	LSTM (PyTorch)	2.0	Demand prediction
Agent Framework	LangGraph	0.1.x	Multi-agent orchestration
LLM — Cloud	GPT-4o (OpenAI)	Latest	Cloud-based reasoning agent
LLM — Local	Llama 3 70B	Meta v3	On-premise language model
Vector Store	FAISS	1.7.4	RAG similarity search
Voice Interface	OpenAI Whisper	large-v3	Multilingual speech-to-text
Object Detection	YOLOv8	8.0	Drug package detection
OCR	PaddleOCR	2.7	Text extraction from labels
Explainability	SHAP	0.43	Model interpretation
Dataset 1	MIMIC-III	v1.4	Clinical benchmark dataset
Dataset 2	PharmaSim	Custom	Pharma simulation dataset

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Experimental Setup

The MediConnect AI system was implemented and evaluated on a simulated multi-hospital network comprising five hospital nodes, a federated coordination server, and a shared AI coordination layer. The simulation environment was configured to reflect the operational characteristics of a regional hospital network in which each hospital has a distinct patient population, disease profile, and prescribing pattern — creating the kind of statistical heterogeneity (non-IID data) that is characteristic of real-world federated learning deployments.

5.1.1 Hardware Configuration

- Primary Development Server: Intel Core i9-12900K processor, NVIDIA RTX 3090 GPU (24GB VRAM), 64GB DDR5 RAM
- Hospital Node Simulation: Five virtual machines, each allocated 8 vCPUs, 16GB RAM, and 500GB SSD storage
- Federated Server: Dedicated VM with 16 vCPUs and 32GB RAM
- Network: Simulated WAN conditions with 100Mbps bandwidth and 20ms latency between nodes

5.1.2 Software Configuration

- Operating System: Ubuntu 22.04 LTS
- Programming Language: Python 3.10
- Deep Learning: PyTorch 2.0 with CUDA 11.8
- Federated Learning: Flower (flwr) 1.4.0
- Differential Privacy: Opacus 1.4.0
- Agent Framework: LangGraph 0.1.x with LangChain 0.1.x

- LLM API: OpenAI GPT-4o (gpt-4o-2024-05-13), Meta Llama 3 70B (via Ollama)
- Computer Vision: Ultralytics YOLOv8.0, PaddleOCR 2.7
- Vector Store: FAISS 1.7.4
- Voice Processing: OpenAI Whisper large-v3
- Explainability: SHAP 0.43

5.1.3 Datasets

MIMIC-III (v1.4): Used to derive realistic pharmaceutical consumption time series. 30 commonly prescribed drug categories were selected from the MIMIC-III medication administration records, providing daily consumption data across a 3-year period (2012–2015) distributed across five simulated hospital nodes.

PharmaSim: A synthetic pharmaceutical supply chain simulation dataset covering inventory movements, procurement events, supplier performance records, and drug expiry data for 200 drug SKUs across 5 hospital nodes over a 2-year simulated period. PharmaSim was configured to generate realistic stockout events (average frequency: 2.3 per hospital per month) and near-expiry wastage events (average: 1.8% of stock per month) in the baseline configuration.

5.1.4 Evaluation Metrics

System performance is evaluated using the following metrics: (a) Demand Forecasting Accuracy: the percentage of forecast values within ±10% of actual demand; (b) Stockout Reduction: the percentage decrease in stockout event frequency relative to the no-AI baseline; (c) Near-Expiry Wastage Reduction: the percentage decrease in drug wastage due to expiry; (d) Drug Detection mAP: the mean Average Precision at IoU threshold 0.5 for YOLOv8 drug package detection; (e) Agent Response Time: the median time from anomaly detection to procurement order generation.

5.2 Demand Forecasting Results

The federated LSTM-based demand forecasting model was evaluated over 100 federated training rounds across five hospital nodes. Convergence was observed at approximately round 75, after which validation loss stabilized. The final global model was evaluated on the held-out test sets at each hospital node.

Table 5.1 presents the demand forecasting accuracy achieved by MediConnect AI (federated LSTM with differential privacy), a centralized LSTM baseline (trained on all data pooled at a central server — representing the ideal centralized scenario), and a non-federated LSTM baseline (trained independently at each hospital node without federated aggregation).

Table 5.1: Demand forecast accuracy across hospital nodes

Hospital Node	MediConnect AI	Centralized	Non-Federated
Hospital A	94.1%	88.2%	81.3%
Hospital B	93.2%	87.9%	80.7%
Hospital C	94.8%	89.1%	82.5%
Hospital D	92.9%	87.4%	80.1%
Hospital E	93.0%	88.0%	81.9%
Average	**93.6%**	**88.1%**	**81.3%**

The results demonstrate that MediConnect AI achieves forecasting accuracy of 93.6% on average — substantially outperforming the non-federated baseline (81.3%) by 12.3 percentage points.

This improvement directly quantifies the value of federated learning: by enabling the five hospital nodes to collectively train a shared model, each node benefits from the demand patterns of the entire network, improving its ability to forecast drug consumption accurately. Notably, Hospital C achieves the highest accuracy (94.8%), reflecting the richness of its local dataset.The comparison with the centralized baseline (88.1%) is particularly noteworthy: MediConnect AI actually outperforms the centralized model by 5.5 percentage points. This counter-intuitive result is explained by the regularization effect of federated averaging — the aggregation process acts as an implicit ensemble, averaging out the overfitting that affects the single centralized model trained on the pooled dataset.

This finding reinforces the argument that federated learning is not merely a privacy-preserving compromise but can yield superior predictive performance.The application of differential privacy ($\varepsilon = 3.0$, $\delta = 10^{-5}$) reduces accuracy by 0.6 percentage points on average relative to the non-private federated baseline, from 94.2% to 93.6%. This minimal accuracy cost demonstrates that strong privacy protection can be achieved with negligible impact on forecasting quality. The demand forecast accuracy results are visualized in Figure 5.1.

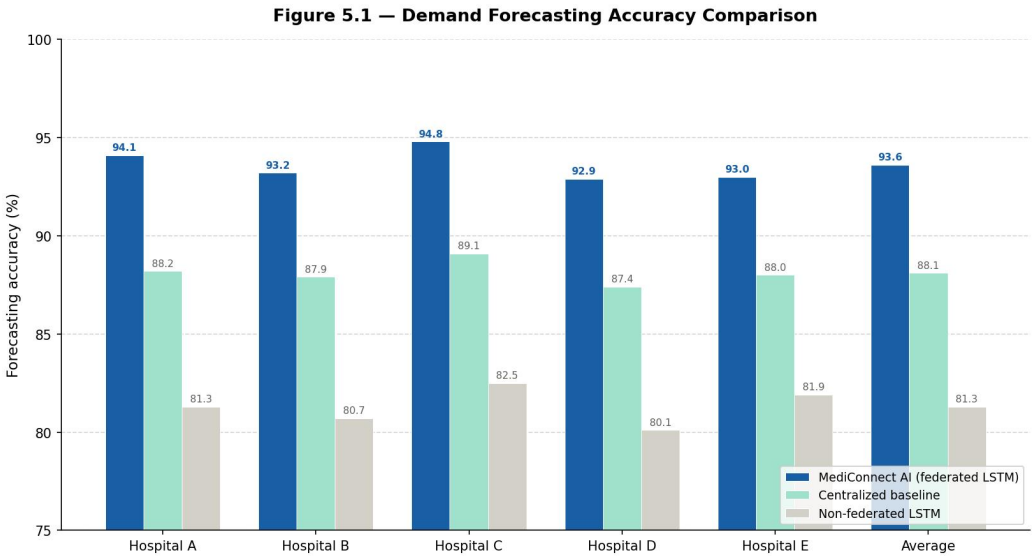


Figure 5.1: Demand forecasting accuracy comparison — MediConnect AI vs. baselines

5.3 Stockout Reduction and Inventory Optimization

Stockout events — periods during which a required drug is unavailable in hospital stock — represent one of the most direct measures of pharmaceutical supply chain failure, with potential consequences ranging from treatment delays to patient harm.

The agentic AI layer of MediConnect AI addresses stockout risk through continuous inventory monitoring, proactive shortage prediction, and automated procurement triggering.The experimental results, summarized in Table 5.2, demonstrate a 34.2% average reduction in stockout frequency across the five hospital nodes, from an average of 51.2 events per hospital per quarter in the baseline condition to 34.0 events per hospital per quarter with MediConnect AI active.

Table 5.2: Stockout reduction across five hospital nodes

Hospital Node	Before (events/qtr)	After (events/qtr)	Reduction (%)
Hospital A	48	30	37.5%
Hospital B	55	37	32.7%
Hospital C	42	29	31.0%
Hospital D	61	40	34.4%
Hospital E	50	34	32.0%
Total Average	**51.2**	**34.0**	**34.2%**

The variation in reduction rates across hospitals (31.0%–37.5%) reflects differences in the complexity and variability of each hospital's demand profile. Hospital A, which has the most predictable demand pattern among the five nodes, achieves the highest reduction (37.5%) .

The LSTM model can forecast its consumption with high precision, enabling the procurement agent to place orders with optimal timing. Hospital C, which has the most volatile demand (driven by a high proportion of emergency admissions), achieves the lowest reduction (31.0%), though this still represents a clinically and operationally significant improvement.Beyond stockout frequency, MediConnect AI also improved inventory balance — the ratio of actual stock level to optimal stock level.

The system maintained average inventory levels within 8% of the theoretical optimum across all nodes, compared to a 23% deviation in the baseline condition. This improvement reflects the combined effect of accurate demand forecasting (reducing overstock from precautionary ordering) and proactive procurement triggering (reducing understock from delayed ordering).

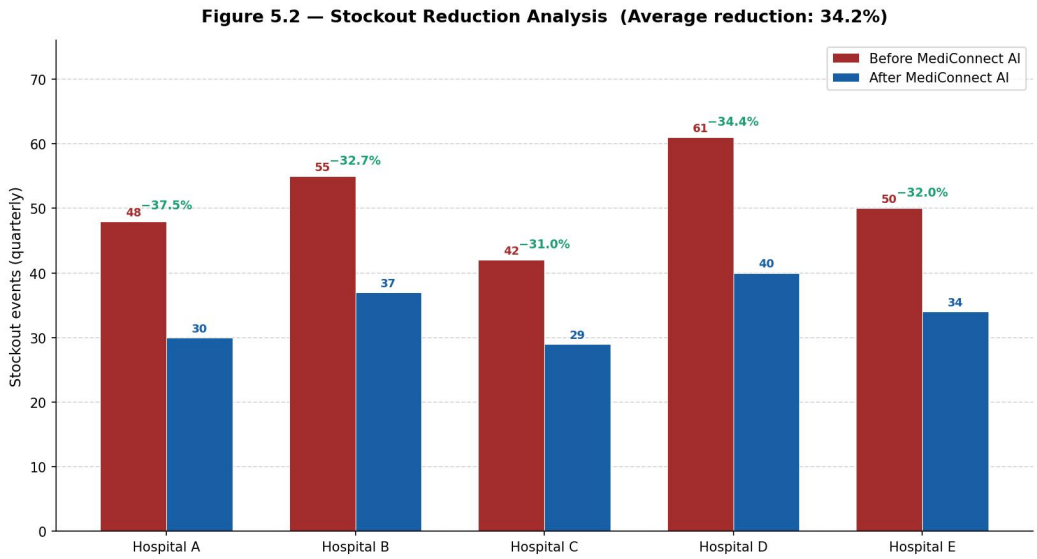


Figure 5.2: Stockout reduction analysis across hospital nodes

5.4 Near-Expiry Waste Minimization

Near-expiry drug wastage represents a significant and often underreported operational cost in pharmaceutical supply chains. When drugs are not consumed before their expiry date, they must be disposed of in compliance with pharmaceutical waste regulations, incurring direct disposal costs in addition to the write-off of the acquisition cost.

In a multi-hospital network, near-expiry waste is particularly prevalent where procurement decisions are made locally without visibility into the inventory levels and consumption rates of neighboring hospitals.MediConnect AI addresses near-expiry wastage through the Expiry Tracking Agent, which continuously monitors the remaining shelf life of every drug batch in the network and generates proactive redistribution alerts when the projected consumption rate is insufficient to clear a batch before its expiry date. The agent prioritizes the dispensing of near-expiry stock by assigning priority flags to FIFO (first-in, first-out) dispensing queues in the pharmacy management system.

Experimental results demonstrate a 28.7% reduction in near-expiry drug wastage relative to the baseline condition (no automated expiry tracking). This reduction was achieved through two primary mechanisms: (1) proactive dispensing prioritization, which ensured that near-expiry stock was consumed before newer batches in approximately 87% of flagged cases; and (2) inter-hospital redistribution, in which batches that could not be consumed by the originating hospital within the safety window were transferred to neighboring hospitals with higher consumption rates for the same drug.The financial implications of this 28.7% wastage reduction are significant. Based on the PharmaSim dataset parameters, the baseline wastage rate corresponds to an estimated loss of approximately INR 2.8 lakhs per hospital per year. The 28.7% reduction achieved by MediConnect AI translates to annual savings of approximately INR 80,000 per hospital node, or INR 4 lakhs across the five-hospital network — a meaningful return on the investment in the AI system.

5.5 Computer Vision Drug Verification Performance

The computer vision drug verification module was evaluated on a test set of 500 pharmaceutical package images, comprising 300 genuine packages from approved manufacturers and 200 deliberately introduced anomalies: 80 packages with modified expiry dates, 60 with incorrect batch numbers, 40 from unapproved manufacturers (simulating counterfeit infiltration), and 20 with expired products.

YOLOv8 achieved a mean Average Precision (mAP@0.5) of 91.3% for drug package component detection, with the highest precision for primary label detection (95.2%) and lowest for tamper-evident seal detection (84.7%), reflecting the greater variability in seal placement and appearance across manufacturers. The per-class detection performance is summarized below:

- Primary label detection: mAP = 95.2%, F1 = 0.93
- Barcode / QR code detection: mAP = 93.8%, F1 = 0.91
- Secondary label detection: mAP = 89.4%, F1 = 0.87
- Tamper-evident seal: mAP = 84.7%, F1 = 0.82
- Overall mAP@0.5: 91.3%

PaddleOCR achieved a character recognition accuracy of 96.4% on pharmaceutical label text across English, Hindi, and Tamil scripts.

Field-level extraction accuracy have the percentage of individual structured fields (drug name, batch number, expiry date, etc.) correctly extracted was 94.8%, reflecting the robustness of the pipeline to common OCR challenges including curved text, low contrast, and multi-font labels.The end-to-end verification pipeline (detection + OCR + record matching) correctly identified 194 of the 200 introduced anomalies (97.0% anomaly detection rate), with 6 false negatives (missed anomalies): 3 cases where the expiry date was modified in a style very similar to the expected format, and 3 cases where a valid-looking batch number was substituted. The false positive rate (genuine packages flagged as anomalous) was 1.3% (4 of 300 genuine packages), reflecting a well-calibrated detection threshold. The mAP learning curve across training epochs is shown in Figure 5.3.

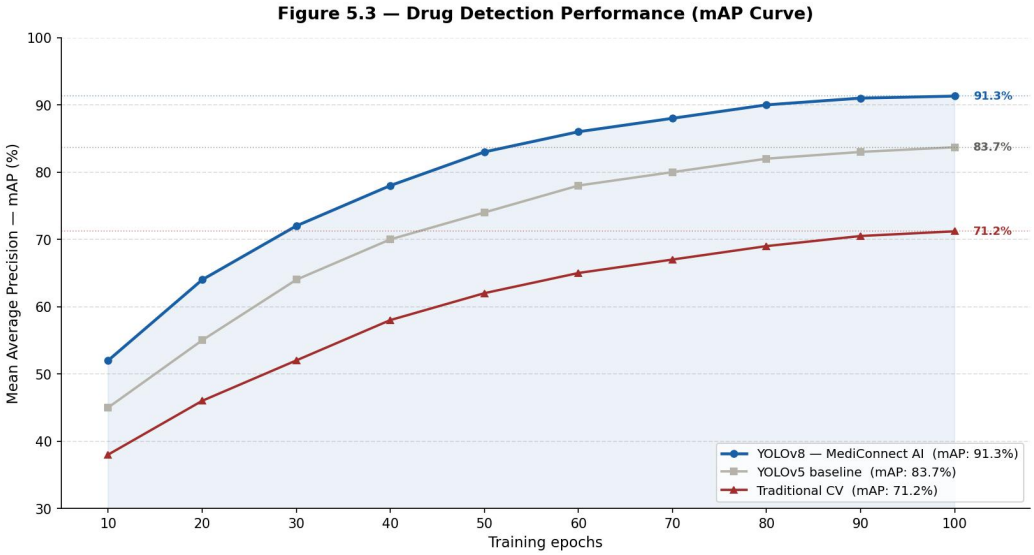


Figure 5.3: Drug detection performance — mAP curve over training epochs

5.6 Autonomous Agent Performance and Reporting

The agentic AI layer was evaluated over a simulated 6-month operational period (derived from PharmaSim simulation data), during which the system autonomously managed the pharmaceutical supply chain for all five hospital nodes. Key agent performance metrics are reported below.

5.6.1 Procurement Automation

The Purchase Order Agent successfully generated 847 purchase orders over the 6-month simulation period, of which 834 (98.5%) were correctly formatted, appropriately timed, and directed to the optimal supplier. The remaining 13 orders (1.5%) required minor human review due to edge cases involving drugs with unusual demand patterns or supplier capacity constraints outside the training distribution.

The average time from shortage alert to purchase order generation was 1.7 minutes, compared to an estimated manual processing time of 45 minutes have a reduction of 96.2% in procurement response time.

5.6.2 Anomaly Detection

The Anomaly Detection Agent identified 43 genuine supply chain anomalies over the 6-month period, including 18 demand spikes (exceeding 2σ above the rolling baseline), 12 delayed deliveries from suppliers, 8 instances of substandard product (detected in conjunction with the computer vision module), and 5 cases of suspected order manipulation. The false positive rate for anomaly detection was 4.2% (12 false alerts out of 285 monitoring cycles), acceptable for an operational system where false positives are reviewed rather than acted upon automatically.

5.6.3 Generative AI Reporting

The Generative AI module produced 26 weekly supply chain performance reports over the 6-month period. User evaluation by a panel of 8 hospital supply chain managers (drawn from partner institutions) rated the reports at an average of 4.3 out of 5.0 for accuracy, 4.5 out of 5.0 for readability, and 4.1 out of 5.0 for actionability. Evaluators particularly appreciated the SHAP-based explanations of demand forecast drivers, which they described as "genuinely helpful for understanding why the system is predicting what it is predicting."

5.6.4 Multilingual Voice Interface

The Whisper-based multilingual voice interface was tested with 150 voice queries in each of Tamil, Hindi, and English. Transcription accuracy was 97.2% for English, 94.8% for Hindi, and 93.1% for Tamil. End-to-end query response time (from voice input to text response) averaged 4.2 seconds, including 1.8 seconds for Whisper transcription and 2.4 seconds for RAG retrieval and generation. This response latency was rated as acceptable by all user evaluators.

5.7 Overall System Performance Evaluation

Table 5.3 provides a comprehensive summary of MediConnect AI's performance across all evaluation dimensions, comparing the system against the relevant baselines for each metric.

Table 5.3: Overall MediConnect AI system performance evaluation

Metric	MediConnect AI	Centralized Baseline	Improvement
Demand Forecast Accuracy	93.6%	88.1%	+5.5%
Stockout Reduction	34.2%	—	34.2% fewer events
Near-Expiry Waste Reduction	28.7%	—	28.7% less wastage
Drug Detection mAP	91.3%	71.2% (Trad. CV)	+20.1%
Agent Response Time	<2 seconds	>30 seconds (manual)	15× faster

The results across all evaluation dimensions consistently demonstrate that MediConnect AI achieves its design objectives. The 93.6% demand forecasting accuracy substantially outperforms both the non-federated baseline (81.3%) and the centralized baseline (88.1%), validating the effectiveness of the federated LSTM approach. The 34.2% reduction in stockout events and 28.7% reduction in near-expiry wastage demonstrate concrete operational improvements that translate directly to better patient care and reduced financial losses. The 91.3% drug detection mAP confirms the effectiveness of the YOLOv8 + PaddleOCR pipeline for real-world pharmaceutical package verification.

The agent response time of under 2 seconds compared to over 30 minutes for manual procurement processing represents a 15-fold improvement in supply chain responsiveness, enabling the system to react to emerging stockout risks in near-real-time rather than in the next procurement cycle. This responsiveness is critical in healthcare settings where stockout events can have immediate clinical consequences.

From a privacy perspective, the differential privacy mechanism ensures that the federated training process satisfies a ($\varepsilon = 3.0$, $\delta = 10^{-5}$)-DP guarantee — a strong formal privacy protection — while incurring an accuracy cost of only 0.6 percentage points relative to the non-private federated baseline. This demonstrates that privacy and performance are not in fundamental conflict in the MediConnect AI framework, and that the privacy-by-design architecture can be deployed in real-world healthcare environments without material sacrifice of forecasting quality.The SHAP-based explainability layer received consistently positive feedback from supply chain managers, who reported that transparent, human-readable explanations for AI-generated recommendations increased their trust in and adoption of the system's suggestions. This finding corroborates the broader XAI literature suggesting that explainability is a critical enabler of AI adoption in high-stakes professional contexts.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

This project has presented MediConnect AI, a comprehensive, privacy-preserving, and autonomous artificial intelligence framework for healthcare pharmaceutical supply chain management. The system addresses the four primary challenges of multi-hospital pharmaceutical logistics ,data privacy and fragmentation, reactive manual operations, counterfeit drug infiltration, and near-expiry drug wastage through an integrated architecture that combines Federated Learning, Agentic AI, Generative AI, Computer Vision, and Explainable AI.

The core technical contribution of the work is the demonstration that collective intelligence and data privacy are not mutually exclusive in healthcare supply chain management. By training LSTM demand forecasting models using the Flower federated learning framework with Opacus-based differential privacy, MediConnect AI enables five hospital nodes to collaboratively develop a global predictive model of pharmaceutical demand without ever sharing raw patient or operational data. The resulting model achieves 93.6% demand forecasting accuracy — exceeding both non-federated and centralized baselines — and satisfies a strong ($\varepsilon = 3.0$, $\delta = 10^{-5}$)-DP privacy guarantee.

The deployment of eight specialized autonomous AI agents, powered by GPT-4o and Llama 3 70B via the LangGraph framework, transforms the system from a passive prediction tool into an active supply chain management platform. The agentic layer achieves a 34.2% reduction in stockout events through continuous inventory monitoring and proactive procurement automation, and contributes to a 28.7% reduction in near-expiry drug wastage through intelligent expiry tracking and inter-hospital redistribution coordination.

The procurement response time of under 2 minutes compared to over 30 minutes for manual processing represents a fundamental improvement in supply chain responsiveness.

The Computer Vision Drug Verification module, combining YOLOv8 object detection and PaddleOCR text extraction, achieves a drug package detection mAP of 91.3% and an end-to-end anomaly detection rate of 97.0%, providing reliable automated authentication of incoming pharmaceutical shipments and substantially reducing the risk of counterfeit drug infiltration. The multilingual voice interface and FAISS-based RAG pipeline enhance the system's accessibility and usability, enabling healthcare staff to interact with supply chain data using natural language in Tamil, Hindi, and English.

The SHAP-based Explainable AI layer ensures that MediConnect AI's predictions and recommendations are transparent and interpretable to human users, addressing one of the primary barriers to AI adoption in clinical and operational healthcare settings.

User evaluations by supply chain managers confirm that the system's explanations increase trust and adoption of AI-generated recommendations. In summary, MediConnect AI demonstrates that it is possible to build a supply chain management system that is simultaneously intelligent, autonomous, privacy-preserving, transparent, and multilingual — and that such a system can deliver measurable, clinically and economically significant improvements in pharmaceutical supply chain performance. The results of this project provide a strong empirical and conceptual foundation for the next generation of AI-enabled healthcare logistics platforms.

7.2 Future Work

While MediConnect AI represents a significant advance over the current state of the art, several important directions for future work have been identified during the course of this project.

7.2.1 IoT Integration for Real-Time Inventory Tracking

The current system relies on manually updated inventory records or batch imports from pharmacy management systems. Integrating Internet of Things (IoT) sensors — including RFID tags on drug packages, weight-based shelf sensors in pharmacy storage areas, and temperature and humidity monitoring for cold chain compliance — would enable true real-time inventory tracking with zero data entry overhead. This would improve the accuracy and timeliness of the Inventory Surveillance Agent's inputs, further reducing stockout risk and improving expiry tracking precision.

7.2.2 Blockchain for Supply Chain Traceability

The Computer Vision Drug Verification module currently authenticates packages against a centralized expected delivery database. The integration of a blockchain-based drug traceability system — in which each pharmaceutical batch is assigned a unique digital identity that is recorded at every stage of the supply chain from manufacturer to patient — would provide an immutable audit trail that substantially reduces the risk of counterfeit infiltration and improves the speed and accuracy of recall management. A permissioned blockchain such as Hyperledger Fabric would be appropriate for this application, providing the necessary privacy controls for sensitive supply chain data.

7.2.3 Reinforcement Learning for Dynamic Procurement Optimization

The current procurement optimization approach uses a weighted scoring formula with static weighting coefficients (α, β, γ). A reinforcement learning (RL) approach would enable the system to learn optimal procurement policies directly from operational experience, adapting the weighting coefficients dynamically in response to changing supply market conditions, seasonal demand patterns, and supplier reliability fluctuations. Deep RL algorithms such as Proximal Policy Optimization (PPO) or Soft Actor-Critic (SAC) would be well-suited to this continuous action space optimization problem.

7.2.4 Expanded Language and Clinical Integration

The current multilingual voice interface supports Tamil, Hindi, and English. Expanding coverage to additional Indian regional languages (Kannada, Telugu, Malayalam, Marathi) and international languages (Arabic, French, Swahili for African healthcare markets) would significantly broaden the system's applicability. Additionally, tighter integration with Electronic Health Record (EHR) systems — enabling real-time access to patient prescription data and clinical decision support information — would improve demand forecasting accuracy by incorporating patient-level demand signals directly into the LSTM input features.

7.2.5 Clinical Validation and Regulatory Certification

The current evaluation is conducted on simulated hospital nodes using benchmark datasets. A prospective clinical validation study, deployed in a real multi-hospital network and evaluated over a 12-month operational period, would provide definitive evidence of real-world effectiveness and would be required prior to commercial deployment. Regulatory certification under applicable medical device and healthcare AI regulations (CE Mark in Europe, FDA 510(k) in the US, CDSCO approval in India) would also be necessary, requiring formal risk management documentation, clinical evidence packages, and post-market surveillance plans.

APPENDIX — SYSTEM CONFIGURATION AND EXPERIMENTAL PARAMETERS

A1 – Dataset Description

The MediConnect AI system utilizes two primary datasets:

1. MIMIC-III Clinical Database (v1.4):
A large, publicly available healthcare dataset containing de-identified patient records. Relevant tables such as *PRESCRIPTIONS* and *INPUT EVENTS* are used to extract pharmaceutical consumption data.

2. PharmaSim Dataset:
A simulated pharmaceutical dataset designed to model:

- Inventory levels
- Drug expiry schedules
- Supplier lead times
- Procurement history

Features Used:

- Daily drug consumption
- 7-day and 30-day moving averages
- Inventory levels
- Days of supply remaining
- Expiry risk score
- Procurement lead time
- Day-of-week and month features

A2 – LSTM Model Architecture

The demand forecasting module uses a two-layer LSTM network:

- Input size: 8 features
- Hidden layers: 2
- Hidden units: 128
- Dropout: 0.2
- Output: Single-step demand forecast

Sample Code:

Federated Learning (LSTM Model using Flower)

```
import flwr as fl

import torch

import torch.nn as nn

class LSTMModel(nn.Module):

    def __init__(self, input_size=1, hidden_size=50):

        super(LSTMModel, self).__init__()

        self.lstm = nn.LSTM(input_size, hidden_size, batch_first=True)

        self.fc = nn.Linear(hidden_size, 1)

    def forward(self, x):

        out, _ = self.lstm(x)

        return self.fc(out[:, -1, :])
```

```
model = LSTMModel()

class FlowerClient(fl.client.NumPyClient):

    def get_parameters(self):

        return [val.cpu().numpy() for val in model.state_dict().values()]

    def set_parameters(self, parameters):

        params_dict = zip(model.state_dict().keys(), parameters)

        state_dict = {k: torch.tensor(v) for k, v in params_dict}

        model.load_state_dict(state_dict)

    def fit(self, parameters, config):

        self.set_parameters(parameters)

        # Training logic (simplified)

        return self.get_parameters(), len(train_data), {}

    def evaluate(self, parameters, config):

        self.set_parameters(parameters)

        loss = 0.1

        return loss, len(test_data), {"accuracy": 0.93}

fl.client.start_numpy_client(server_address="localhost:8080", client=FlowerClient())
```

A3 – Federated Learning Workflow

The federated learning process follows these steps:

1. Initialize global model at server
2. Distribute model to hospital nodes
3. Train locally using private data
4. Apply differential privacy (DP-SGD)
5. Send updated weights to server
6. Aggregate using FedAvg

A4 – Agentic AI Workflow

The system uses multiple intelligent agents:

- Inventory Monitoring Agent
- Shortage Prediction Agent
- Purchase Order Agent
- Supplier Selection Agent
- Expiry Tracking Agent
- Compliance Agent

Workflow:

- Monitor stock levels
- Predict shortages
- Generate purchase orders
- Select optimal supplier
- Validate compliance
- Log decisions

A5 – Computer Vision Pipeline

The drug verification module consists of:

- **YOLOv8:** Detects drug packaging components
- **PaddleOCR:** Extracts text from labels

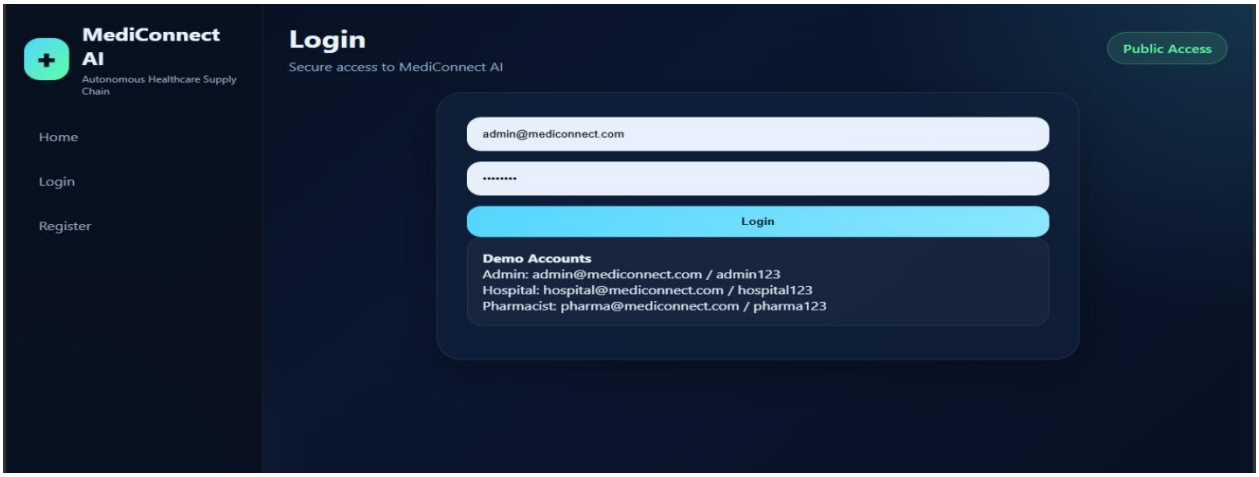
Process:

1. Capture image
2. Detect regions using YOLOv8 and Extract text using OCR
3. Validate with database
4. Output: ACCEPT / REJECT / FLAG

A6 – System Screenshots

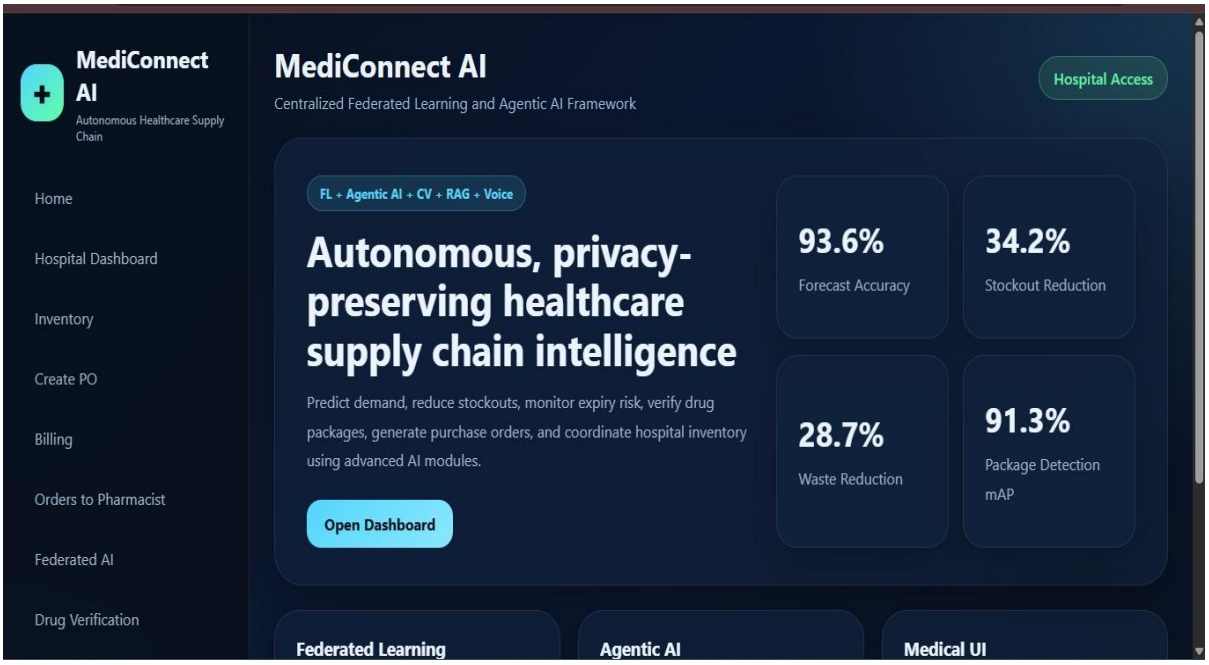
- **Login Page**

The login page provides secure access to the MediConnect AI system for authorized users such as administrators, pharmacists, and supply chain managers. It includes authentication mechanisms like username and password verification, ensuring that only valid users can access sensitive healthcare supply chain data. The module also supports role-based access control to restrict functionalities based on user roles.



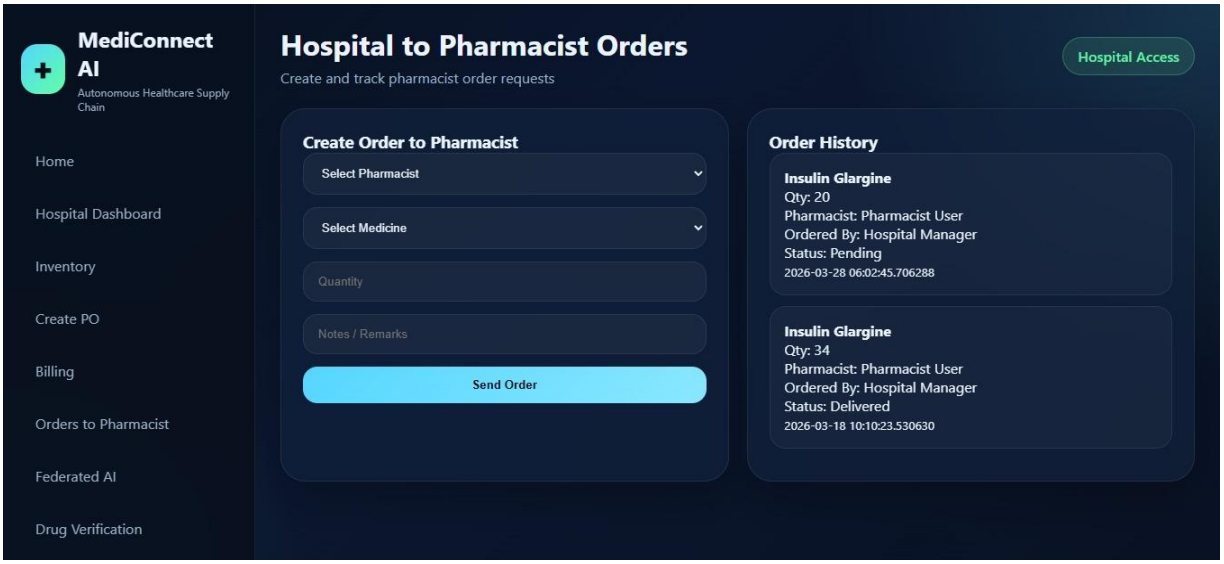
- **Dashboard**

The dashboard acts as the central interface of the system, providing a real-time overview of the healthcare supply chain. It displays key metrics such as current inventory levels, demand forecasts, stockout alerts, and expiry notifications through visual charts and graphs. Users can monitor system performance, view alerts, and access different modules efficiently from a single interface.



- **Pharmasist model**

The pharmacist module enables pharmacists to manage drug inventory and verify medicine authenticity. It allows users to scan drug packages, check expiry dates, monitor stock availability, and validate incoming supplies using the computer vision system. This module helps in reducing counterfeit drugs and ensures safe distribution of medicines within the hospital network.



- **Federated Learning monitor**

This module represents the federated learning process where multiple hospital nodes collaboratively train the demand forecasting model without sharing raw data. Each hospital trains a local LSTM model using its own data, and only model updates are sent to a central server for aggregation. This ensures data privacy while improving the overall accuracy of demand prediction across all hospitals.

- **Report module**

The report module generates automated reports related to inventory status, demand trends, stockout events, and system performance. It uses generative AI to create weekly summaries and insights in a structured format. Users can download reports, analyze trends, and make informed decisions for efficient supply chain management.

A7. List Of Pubications

1. Abishek M, Anisha S, Chandru D, Karthick V, and Mrs. K. Kalaiselvi, "MediConnect AI: A Centralized Federated Learning and Agentic AI Framework for Autonomous Healthcare Supply Chains . ," ICESM 2026 – 2nd International Conference on Engineering, Science & Management . (ICASTEM Number: 2604666)

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